
Causal Inference

DRP

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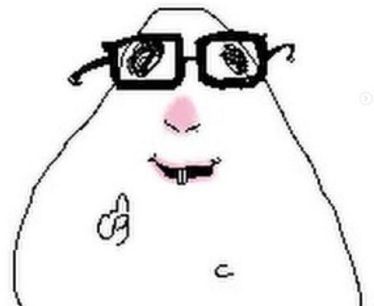
Summary

Textbook: “A First Course in Causal Inference” by Peng Ding

- Chapter 1, parts of chapter 2 and 3

TOPICS

1. Introduction
2. Yule-Simpson Paradox
3. Potential Outcomes
4. The CRE and the Fisher Randomization Test
5. Neyman Repeated Sampling Inference



Introduction

“Correlation does not imply causation.”



- In classical statistics focuses on association and not causation
- Causal inference aims to estimate causal and effect relationships in experiments and observational data

Yule-Simpson Paradox

> Trends within groups contradict the overall trend.

Example: UC Berkeley Admissions Data

- Overall percentage of men that were accepted into the school was significantly more than the percentage of women
- But in several departments the percentage of women admitted was much higher!
- Women were more likely to apply to more competitive departments than men, resulting in this paradox

Men		Women	
Applicants	Admitted	Applicants	Admitted
8,442	44%	4,321	35%

Men		Women	
Applicants	Admitted	Applicants	Admitted
825	62%	108	82%
560	63%	25	68%

Department A >
Department B >

Potential Outcomes Model

> Model assumptions used to quantify causal effects. Addresses the Yule-Simpson paradox.

- Two treatment groups (typically treatment=1, or no treatment=0)
- (Consistency): when a unit receives no treatment or treatment, why will exhibit the potential outcome (Outcomes with and without treatment, both for each unit are assumed to be fixed)*
- (No interference): Treatment of one unit does not impact the outcome of another

NOTATION:

- i = index of unit (e.g. $i = 1$ = patient #1)
- $Y_i(0)$ = potential outcome of i given no treatment
- $Y_i(1)$ = potential outcome of i given treatment

* In reality, **we will only see $Y_i(0)$ OR $Y_i(1)$** for a given unit i (test subjects are only given treatment or no treatment)

POTENTIAL OUTCOMES

i	$Y_i(1)$	$Y_i(0)$
1	$Y_1(1)$	$Y_1(0)$
2	$Y_2(1)$	$Y_2(0)$
$\vdots$	$\vdots$	$\vdots$
n	$Y_n(1)$	$Y_n(0)$

- n number of units
- Each has a fixed outcome with (1) and without treatment (0)

INDIVIDUAL CAUSAL EFFECTS

$$\tau_i = Y_i(1) - Y_i(0), \quad (i = 1, \dots, n)$$

This is challenging to observe because a given unit only gets one treatment.

$$\tau = n^{-1} \sum_{i=1}^n \{Y_i(1) - Y_i(0)\}$$

So we focus on mean causal effect instead.

Completely Randomized Experiment (CRE)

Key Points:

- n_1 (# of units receiving treatment) and n_0 (# of units receiving no treatment) are fixed
- $n_1 + n_0 = n$
- Let Z be vector of length n , where each z_i denotes whether the unit at index i gets treatment ($z_i = 1$) or not ($z_i = 0$)

Thus:

- There are $\binom{n}{n_1}$ number of possible Z (assignment vectors)
- These possible Z are uniformly distributed (all equally likely)

Fischer Randomized Test

Null Hypothesis: For any given unit, its outcome under the treatment will equal its outcome given no treatment

$$Y_i(1) = Y_i(0) \quad \text{for all } i = 1, 2, \dots, n$$

- Any test statistic works under the FRT (some are more intuitive than others)
- Let $\mathbf{Y} = Y(1) = Y(0)$ (which is a vector of sample outcomes)
- If the null hypothesis is true, this means our only source of randomness is Z , the assignment vector

Choice of Test Statistic

Difference in means statistic:

$$\hat{\tau} = \hat{Y}(1) - \hat{Y}(0)$$

- Sample mean of treatment outcomes - sample mean of none treatment outcomes
- Under the null hypothesis, $E[\tau_hat] = 0$

P-Value by Fischer (True P-Value)

M = number of possible Z

I = indicator function that returns 1 if the test statistic under z^m is equally or as extreme as the sample, 0 otherwise

$$M^{-1} \sum_{m=1}^M I\{T(z^m, \mathbf{Y}) \geq T(\mathbf{Z}, \mathbf{Y})\}$$

- Recall that Z is uniformly distributed, with n choose n1 possible values
- As n increases (meaning we have a large sample), the possible Z's get incredibly large
 - Example: with n = 100, and n1 = 50, we have over 10^{29} possible Z!
- Calculating the above equation then becomes computationally difficult
- **Returns: Proportion of all possible assignments that result in a test statistic more or equally as extreme as the sample**

P-Value by Monte Carlo

R = some large, but feasible number (like 1,000 or 10,000)

I = indicator function that returns 1 if the test statistic under z^r is equally or as extreme as the sample, 0 otherwise

$$R^{-1} \sum_{r=1}^R I\{T(z^r, Y) \geq T(Z, Y)\}$$

- We now only have R values of Z to consider
- Computationally feasible even for very large n
- **Returns: Proportion of a feasible subset of possible assignments that result in a test statistic more or equally as extreme as the sample**
- (The indicator function is $\geq$ because we are looking at causal effect)

Example using LaLonde Data

*LaLonde data: outcomes after job training vs no job training

```
library(Matching)
data ( lalonde )
z = lalonde $ treat
y = lalonde $ re78

# z = assignment
# y = outcome

# run test, tau_hat = mean of Y1 - Y0
tau_hat <- (sum(y[z==1])/sum(z==1)) - (sum(y[z==0])/sum(z==0))

t.test(y[z==1], y[z==0], var.equal = TRUE)$statistic

# run my own monte carlo p-value test
R <- 10^4
p_val <- 0
for(i in 1:R){
  q <- sample(z, size=length(z), replace=F)
  if((sum(y[q==1])/sum(q==1)) - (sum(y[q==0])/sum(q==0)) >= tau_hat){
    p_val = p_val + 1
  }
}

p_val <- p_val / R
p_val
```

< get Y and Z

< calculate tau_hat

< get monte carlo p-value using R = 10,000

Results: After running several times, it was always well below 0.01

Neyman's Inference

$$\hat{\tau} = \hat{Y}(1) - \hat{Y}(0)$$

* This is the difference in means test statistic

> Under a CRE, the difference in means test statistic is an unbiased estimator of tau (the true mean causal effect).

$$E(\hat{\tau}) = \tau;$$

$$\begin{aligned}\text{var}(\hat{\tau}) &= \frac{S^2(1)}{n_1} + \frac{S^2(0)}{n_0} - \frac{S^2(\tau)}{n} \\ &= \frac{n_0}{n_1 n} S^2(1) + \frac{n_1}{n_0 n} S^2(0) + \frac{2}{n} S(1, 0);\end{aligned}$$

- $S^2(1)$ = sample variance of $Y(1)$
- $S^2(0)$ = sample variance of $Y(0)$
- $S(1,0)$ = sample covariance

Applications to Observational Studies

> It is fundamentally difficult to directly apply causal techniques to observational studies because the researcher does not control the assignment groups (assignment is not random).

- Conditioning on X , the possible confounding variables
- Extra assumptions based on X

Thank you for listening!

